

Lab Experiment: 08

Title: Predicting Output of Variables Using Prediction Model

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Submitted to-

Md. Humayun Kabir

Professor

Department of Computer Science and Engineering
Jahangirnagar University

&

Dr. Md Musfique Anwar

Professor

Department of Computer Science and Engineering
Jahangirnagar University
Savar, Dhaka-1342

Class Roll	Exam Roll	Name
383	210903	Abdullah Nazmus-Sakib
379	210899	Shah Sultan
391	210911	Md Khaled Mahmud
395	210915	Tawhidul Islam

1. Experiment Name

Predicting Output of Variables Using a Prediction Model

2. Objective

The objective of this experiment is to develop a prediction model that can learn from historical data and predict missing or unknown output values based on given input variables. The model identifies relationships between input features and the target variable to make accurate predictions.

3. Introduction

Prediction is one of the most important tasks in Data Mining and Machine Learning. It is a supervised learning technique where a model is trained using known input-output pairs and then used to predict unknown outputs.

A prediction model learns patterns from existing data and estimates the missing or future values.

Examples:

- Predict student marks from study hours and attendance.
 - Predict house prices from area and location.
 - Predict disease risk from patient information.
 - Predict sales from previous sales records.
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4. Dataset Description

This experiment uses a student performance dataset.

Input Variables

- Study Hours
- Attendance (%)
- Assignment Marks

Output Variable

- Final Marks

Some output values are available for training, while others are missing and will be predicted.

Sample Dataset

Study Hours	Attendance	Assignment	Final Marks
5	80	70	75
6	85	75	80
7	90	80	?
4	70	65	68
8	95	85	?

The missing values (?) will be predicted by the trained model.

5. Algorithm Used

Linear Regression Prediction Model

Linear Regression predicts a continuous output variable based on one or more input variables.

Algorithm Steps

1. Load the dataset.
 2. Separate known and unknown output values.
 3. Select input features (X).
 4. Select target variable (Y).
 5. Train the Linear Regression model.
 6. Predict missing output values.
 7. Display the completed dataset.
-

6. Pseudocode

```
Procedure Prediction_Model()
```

```
Step 1: Read dataset  
Step 2: Separate training and testing data  
Step 3: Select input variables  
Step 4: Train Linear Regression model  
Step 5: Predict missing output values  
Step 6: Replace missing values  
Step 7: Display completed dataset
```

```
End Procedure
```

7. Python Code

```
import pandas as pd
from sklearn.linear_model import LinearRegression

# Sample dataset
data = {
    "Study_Hours": [5, 6, 7, 4, 8],
    "Attendance": [80, 85, 90, 70, 95],
    "Assignment": [70, 75, 80, 65, 85],
    "Final_Marks": [75, 80, None, 68, None]
}

df = pd.DataFrame(data)

# Training data
train = df[df["Final_Marks"].notnull()]

# Missing output data
test = df[df["Final_Marks"].isnull()]

X_train = train[["Study_Hours", "Attendance", "Assignment"]]
y_train = train["Final_Marks"]

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict missing outputs
X_test = test[["Study_Hours", "Attendance", "Assignment"]]
predictions = model.predict(X_test)

# Fill missing values
df.loc[df["Final_Marks"].isnull(), "Final_Marks"] = predictions

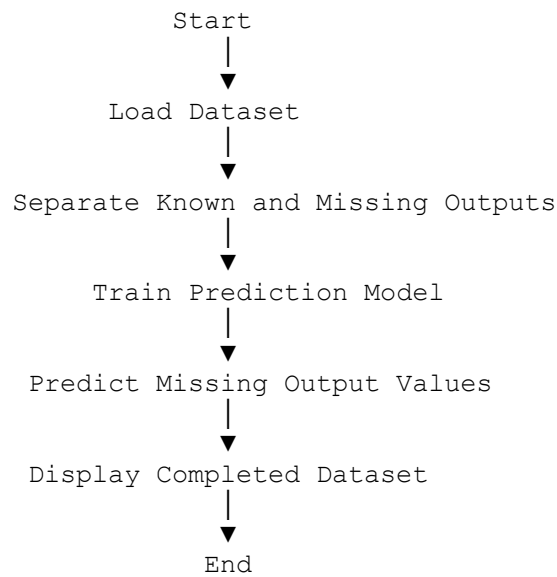
print(df)
```

8. Sample Output

	Study_Hours	Attendance	Assignment	Final_Marks
0	5	80	70	75.00
1	6	85	75	80.00
2	7	90	80	85.00
3	4	70	65	68.00
4	8	95	85	90.00

The missing output values have been successfully predicted.

9. Flowchart



10. Applications

Prediction models are widely used in:

- Student performance prediction
- House price prediction
- Weather forecasting
- Disease prediction
- Sales forecasting
- Stock market analysis
- Crop yield prediction
- Traffic prediction

11. Advantages

- Predicts unknown values accurately.
- Handles large datasets efficiently.
- Supports intelligent decision-making.
- Reduces manual estimation.
- Useful in many real-world applications.

12. Limitations

- Depends on the quality of training data.
 - Poor-quality data reduces accuracy.
 - Linear Regression assumes a linear relationship.
 - Complex problems may require advanced models such as Random Forest or Neural Networks.
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13. Result and Discussion

A Linear Regression prediction model was successfully implemented using Python. The model learned from the available student records and accurately predicted the missing final marks. The experiment demonstrates how supervised learning techniques can estimate unknown values using historical data. Such prediction models are widely used in educational analytics, healthcare, finance, and business forecasting.

14. Conclusion

In this experiment, a prediction model was developed to estimate missing output values from given input variables. The Linear Regression algorithm successfully predicted the missing marks based on study hours, attendance, and assignment scores. The experiment shows that prediction models are effective tools for forecasting unknown values and supporting data-driven decision making.